REST Countries — Milestone 1 Report

Data profiling, cleaning decisions and validation, exploratory visualizations, and references

Scope: 254 country and territory records from the REST Countries API. Profiling results are identified as pre-cleaning; subsequent cleaning actions are documented separately.

1. Data Profiling — pre-cleaning findings

This section reproduces the findings in the supplied profiling report, before cleaning or enrichment. Empty-string checks and null-value checks are separate and should not be conflated.

Data overview

Column	Type / representation
CountryName	Object (text)
Region	Object (text)
Area(km2)	Float; square kilometers
Languages	Comma-separated language names
CurrencyCodes	Comma-separated currency codes
borders	Comma-separated land-neighbor codes
Population	Integer count

Completeness checks (pre-cleaning)

Check	Finding	% of 254
Languages: empty string	1	0.39%
CurrencyCodes: empty string	3	1.18%
borders: empty string	86	33.86%
CountryName, Region, Area(km2), Population: null	0 each	0.00% each

Empty strings were measured with **df[column] == ""** for the three text-list fields. Nulls in the other four fields were measured with **isna().sum()**. A blank borders field can describe an island or territory with no land neighbor.

1. Data Profiling — pre-cleaning findings (continued)

Descriptive statistics

Statistic	Area (km²)	Population
Count	254	254
Mean	591,731.2	31,779,170
Standard deviation	1,895,959	131,409,200
Minimum	0.44	0
Q1 (25%)	1,439.75	222,007.8
Median (50%)	62,979	4,920,816
Q3 (75%)	353,423.8	20,540,570
Maximum	17,098,240	1,429,404,000

Values retain the displayed precision from the supplied pandas describe() output. In both numeric variables, the means exceed the medians, consistent with right-skewed distributions.

Outlier screening (1.5 × IQR rule)

Numeric column	Flagged	% of 254
Area(km2)	35	13.78%
Population	30	11.81%

Flags do not imply errors, and the counts cannot be added to calculate unique flagged records because they may overlap.

Records by API region

Region	Records	% of 254
Africa	60	23.62%
Americas	56	22.05%
Europe	54	21.26%
Asia	52	20.47%
Oceania	27	10.63%
Antarctic	5	1.97%
Total	254	100.00%

1. Data Profiling — pre-cleaning findings (continued)

Duplicate and validity checks

Check	Pre-cleaning finding
Duplicate full rows	0
Area ≤ 0	0
Population < 0	0
Population = 0	2 records: Bouvet Island; Heard Island and McDonald Islands

Main interpretation

The checked numeric fields are complete, and no duplicate full rows, nonpositive areas, or negative populations were found. The largest number of blank strings is in borders; this often represents genuine absence of land neighbors rather than a data-quality defect. Substantial numeric spread and IQR flags warrant review, not automatic deletion.

Percentages in Section 1 are calculated from the supplied counts using n = 254 and rounded to two decimal places. No post-cleaning numeric summary statistics were supplied.

2. Cleaning Decisions and Validation

Cleaning retained 254 records. The raw dataset was kept separate from the cleaned output. The following decisions are documented in the supplied cleaning notes; linked materials are listed in Section 4.

Language and currency fields

Record / field	Decision	Reason
Antarctica / Languages	Keep blank	No single continent-wide official or universally adopted language was identified.
Antarctica / CurrencyCodes	Keep blank	The Australian Border Force reference lists no universal currency for Antarctica.
Bouvet Island / CurrencyCodes	Set NOK	Territory-associated currency code from Australian Border Force table.
Heard Island and McDonald Islands / CurrencyCodes	Set AUD	Territory-associated currency code from Australian Border Force table.

2. Cleaning Decisions and Validation (continued)

Borders: verified additions and special cases

Record	Cleaned action	Rationale
Greenland	Add CAN	Canada–Denmark/Greenland 2022 Hans Island (Tartupaluk) agreement.
Cyprus	Add GBR	UK Sovereign Base Areas (Akrotiri and Dhekelia); not mainland UK.
Åland Islands	Add SWE	The Finland–Sweden land border crosses Märket island.
Canada	Add GRL reciprocally	Consistency with the Greenland–Canada land border.
Cuba	Keep blank	Guantánamo Bay naval base boundary is excluded under the sovereign-country-border definition.
Antarctica	Keep blank	Territorial claims do not establish universally accepted sovereign borders.
French Southern and Antarctic Lands	Keep blank	Antarctic territorial claims do not establish universally accepted sovereign borders; classification remains a special case.

Border completeness validation

Measure	Pre-cleaning	After cleaning
All records	254	254
Blank borders strings	86 (33.86%)	83 (32.68%)

The three previously blank records corrected were Greenland, Cyprus, and Åland Islands. Canada also received reciprocal GRL; it was not one of the blank-border records. The blank-border count fell by three, from 86 to 83.

Population and numeric outliers

Zero population: Retained both Bouvet Island and Heard Island and McDonald Islands, consistent with their uninhabited status in the supplied notes. A valid zero should not be imputed or removed solely because it is zero.

IQR-flagged values: Retained all 35 area flags and 30 population flags. The team's review did not identify an error warranting correction; outlier status alone is not evidence of invalidity.

Validation boundaries

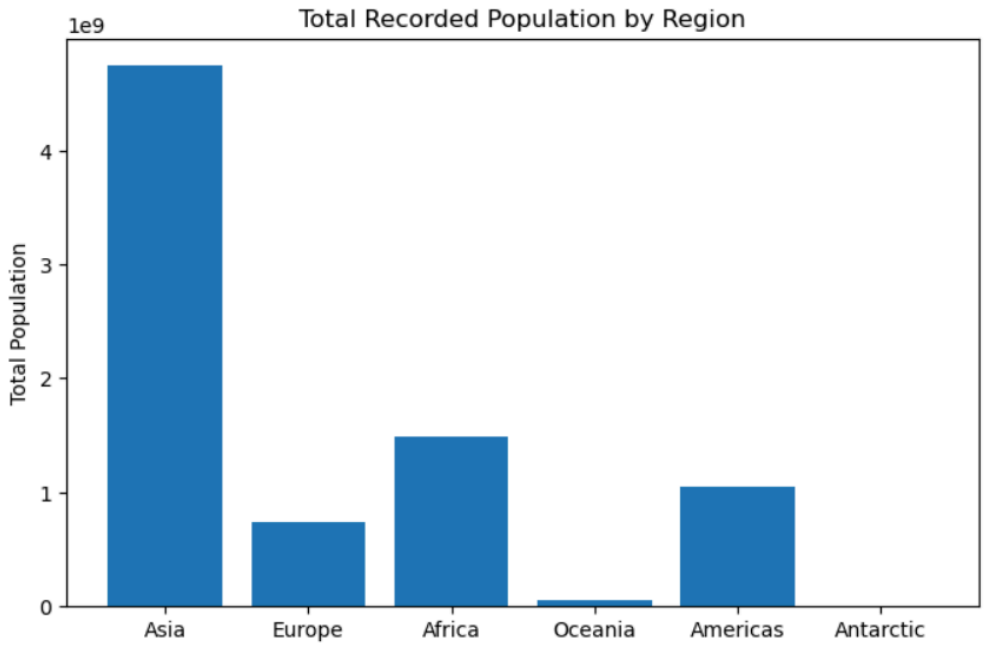
The provided cleaning notes support a final row count of 254 and 83 blank borders fields. They do not supply updated counts for all other fields, so this report does not claim revised missing-value totals, regional totals, or descriptive statistics after cleaning.

3. Exploratory Visualizations

Five original figures are presented in the requested order, with the analytical questions and supplied interpretations.

Figure 1. Total recorded population by region

Analytical question: Which region has the highest total population?

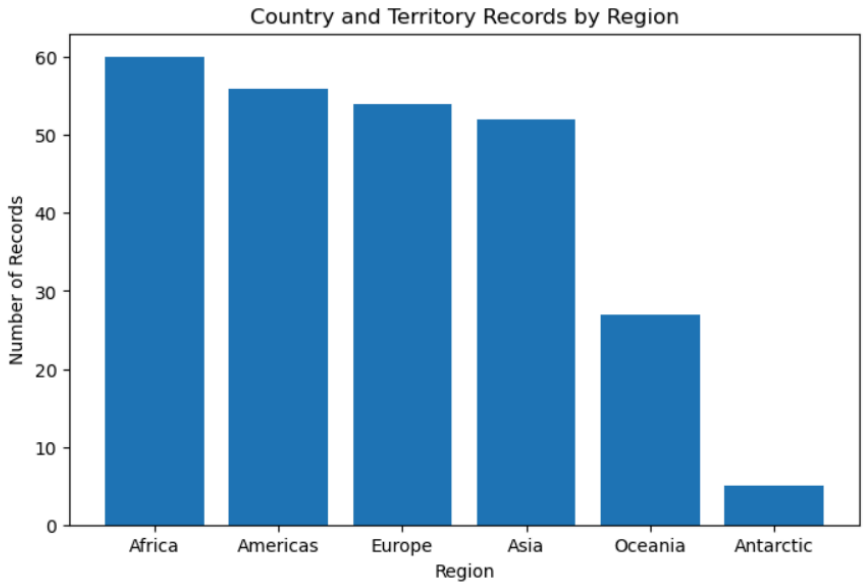


Interpretation

Asia has the highest total recorded population in the dataset, at approximately 4.75 billion, followed by Africa at 1.49 billion.

Figure 2. Country and territory records by region

Analytical question: Which region has the highest number of records?

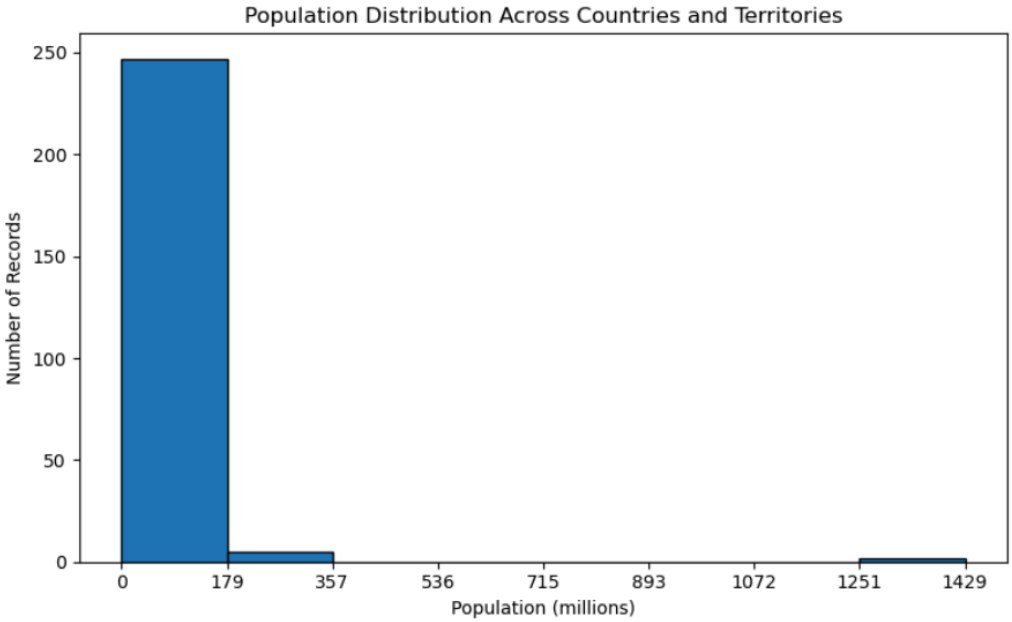


Interpretation

This chart shows the number of country and territory records in each region. Africa has the most records (60), followed by the Americas (56), while the Antarctic region has the fewest (5).

Figure 3. Population distribution across countries and territories

Analytical question: How are the population sizes of countries and territories distributed in the dataset?

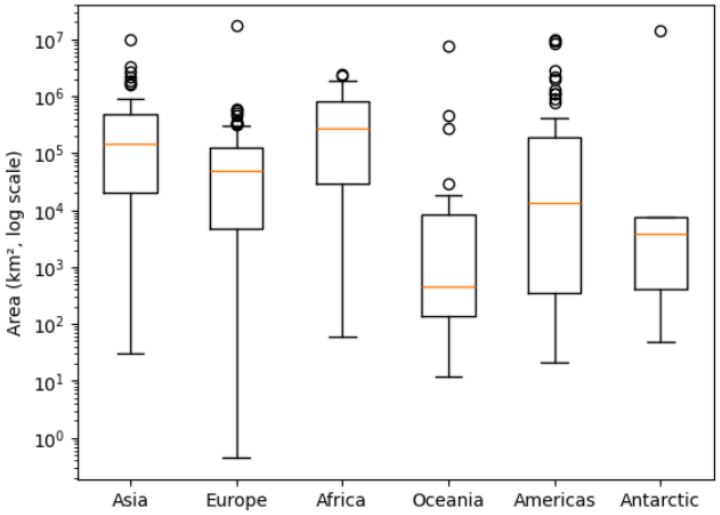


Interpretation

The histogram groups countries and territories by population size. Of the 254 records, 247 fall in the first interval, from 0 to about 178.68 million people. Only two records fall in the highest interval, from about 1.25 to 1.43 billion people.

Figure 4. Area boxplot by region

Analytical question: How do the median and spread of country and territory areas vary by region?

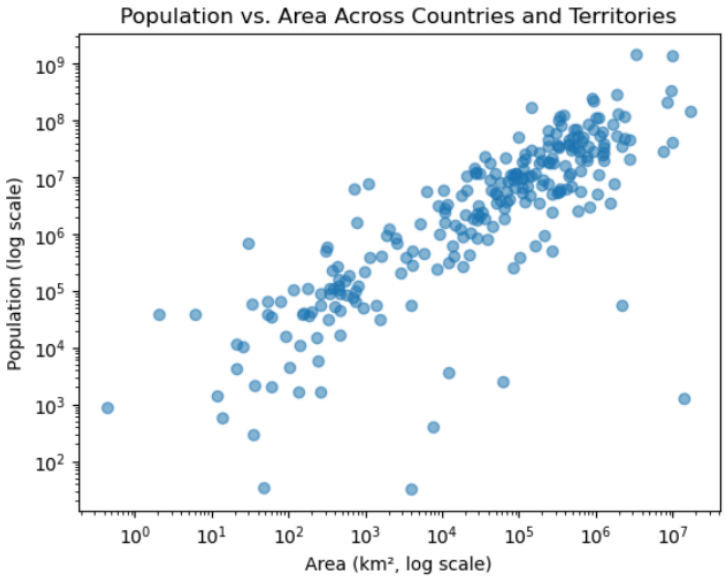


Interpretation

Each box shows the middle 50% of country and territory areas in a region, and the orange line marks the median. Africa has the highest median area, while Oceania has the lowest. The Americas have the widest box on the logarithmic scale, showing the largest relative spread between the first and third quartiles.

Figure 5. Population vs. area across countries and territories

Analytical question: Do countries and territories with larger areas tend to have larger populations?



Interpretation

Each point represents a country or territory. The overall upward pattern shows that larger areas tend to be associated with larger populations, although records with similar areas can have very different population sizes. Both axes use a logarithmic scale. Two records with zero population were excluded from the plot because zero cannot be displayed on the logarithmic population axis.

Axis note: Area (km²) and population are on logarithmic axes. The two zero-population records remain in the dataset but are excluded from this scatter plot.

4. References

The following links are drawn from the supplied cleaning notes or provide context for the special Antarctic cases. They are presented for review; the cited websites were not independently rechecked while assembling this PDF.

Sources

1. REST Countries API
https://restcountries.com/

2. Australian Border Force — Invoice currency code list
https://www.abf.gov.au/help-and-support/ics/integrated-cargo-system-%28ics%29/software-developers/reference-materials/code-list/invoice-currency-code

3. Government of Canada — Boundary dispute
https://www.canada.ca/en/global-affairs/news/2022/06/boundary-dispute.html

4. Government of Canada — Canada/Denmark/Greenland historic agreement
https://www.canada.ca/en/global-affairs/news/2022/06/canada-and-the-kingdom-of-denmark-together-with-greenland-reach-historic-agreement-on-long-standing-boundary-disputes.html

5. CIA World Factbook archive — Cyprus land boundaries
https://www.thefactbook.org/cyprus/land-boundaries

6. UK House of Commons Library — Sovereign Base Areas
https://commonslibrary.parliament.uk/research-briefings/cbp-10540/

7. Nordic cooperation — About Åland
https://www.norden.org/en/information/about-aland

8. Swedish mapping authority — Finland/Sweden boundary
https://www.lantmateriet.se/sv/om-lantmateriet/Samverkan-med-andra/sveriges-riksgranser/riksgransen-sverige-finland/

9. Australian Government — Heard marine habitats report
https://www.agriculture.gov.au/sites/default/files/documents/heard-marine-habitats.pdf

10. Australian Antarctic Program — Heard Island
https://www.antarctica.gov.au/antarctic-operations/stations-and-field-locations/heard-island/

11. Antarctic Treaty Secretariat
https://www.ats.aq/e/antarctictreaty.html

Report inputs: supplied REST Countries Data Profiling Report (pre-cleaning), cleaning.txt notes, and figure screenshots/captions. All numerical values are from these materials; percentages are computed from 254 records and rounded to two decimal places.